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# LogSync: A Low-Power Logarithmic SoC for Cross-Regional Synchrony Extraction and Adaptive Psychiatric State Classification

Daney Alex, *Student Member, IEEE*, Amitabh Yadav, *Student Member, IEEE*, Jieun Joo, *Student Member, IEEE*, Uisub Shin, *Student Member, IEEE*, Jinhan Liu, Geoffrey W. Diehl, Volkan Cevher, *Fellow, IEEE*, Alik S. Widge, and Mahsa Shoaran, *Senior Member, IEEE*

***Abstract*—Closed-loop deep brain stimulation (DBS) has emerged as a promising therapeutic approach for treatment-resistant psychiatric disorders, offering the potential to modulate pathological network activity in real time. To support such interventions, we present a 32-channel brain–computer interface (BCI) system-on-chip (SoC) that integrates two key functions: 1) adaptive, biomarker-driven classification of psychiatric states and 2) phase-locked closed-loop DBS. The SoC incorporates a 32-channel low-noise analog front-end (AFE), a logarithmic arithmetic unit (LAU) for efficient and accurate feature extraction (FE), including spectral power, phase, and connectivity biomarkers, a neural additive model (NAM) classifier with backpropagation-free online adaptation, and a programmable multi-mode stimulation controller that enables flexible coordination with an external stimulator. Fabricated in 65-nm CMOS, the chip occupies 5.46 mm$^2$ and consumes 6.14 $\mu$W/channel while achieving an energy efficiency of 0.197 $\mu$J/classification. The proposed SoC was extensively validated on two intracranial electroencephalography (iEEG) datasets related to psychiatric disorders, achieving 80.9%/74.8% sensitivity and 74.2%/73.1% specificity on the multi-source interference task (MSIT) and the emotional conflict resolution (ECR) task, respectively. We further evaluated the SoC on the public DEAP dataset for electroencephalography (EEG)-based emotion recognition, achieving 70.5%/72.5% sensitivity and 69.8%/66.3% specificity for the arousal and valence states, respectively. In addition, we performed in vitro measurements and in vivo characterization in a Long-Evans rat to validate the AFE and phase-locked stimulation controller, confirming the feasibility of the proposed SoC for real-time intervention. Overall, the architecture enables accurate, low-power, and interpretable classification of complex neural states, supporting adaptive DBS for psychiatric neuromodulation and other BCI applications.**

***Index Terms*—Knowledge distillation, logarithmic number system (LNS), neural additive model (NAM), neural connectivity biomarkers, phase-locked stimulation, psychiatric disorders.**

## I. Introduction

PSYCHIATRIC disorders are among the most pressing global health challenges, affecting more than 970 million people worldwide [1]. Depression and anxiety disorders alone affect more than 581 million people. Although pharmacological and behavioral therapies have seen progress, a substantial proportion of patients with neuropsychiatric conditions fail to achieve meaningful or lasting relief, motivating the search for alternative neuromodulation-based interventions [2], [3], [4], [5].

From a therapeutic perspective, deep brain stimulation (DBS) has emerged as a promising treatment modality for neurological disorders [4], [7], [8]. Traditional open-loop DBS, typically delivered at high frequencies, has shown significant efficacy in alleviating symptoms of movement disorders such as Parkinson's disease (PD) and essential tremor [9], [10]. More recently, its applications have expanded to a wider range of neurological and psychiatric conditions, including epilepsy, stroke rehabilitation, depression, and memory-related dysfunctions [2]. Despite its clinical benefits, open-loop DBS is often associated with several limitations, including undesirable side effects, high power consumption, frequent battery replacements, and reduced long-term efficacy [11], [12]. Recent studies suggest that closed-loop DBS, driven by neural biomarkers [12], [13], [14], [15], [16], can be effective in managing movement disorders and also shows emerging promise in the treatment of psychiatric conditions such as major depression [8], [17]. In particular, recent findings indicate that closed-loop, *phase-locked* DBS [14], [15] can match or even surpass the efficacy of conventional open-loop stimulation in movement disorders [16] and may open a particularly promising avenue for treating psychiatric illnesses. By delivering stimulation bursts precisely aligned with the

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Daney Alex, Amitabh Yadav, Jieun Joo, Jinhan Liu, and Mahsa Shoaran are with the Institutes of Electrical and Micro Engineering and Neuro-X, École Polytechnique Fédérale de Lausanne, 1202 Geneva, Switzerland (e-mail: daney.alex@epfl.ch; amitabh.yadav@epfl.ch).

Uisub Shin was with the Institutes of Electrical and Micro Engineering and Neuro-X, École Polytechnique Fédérale de Lausanne, 1202 Geneva, Switzerland, and also with the School of Electrical and Computer Engineering, Cornell University, Ithaca, NY 14853 USA. He is now with Apple Inc., Sunnyvale, CA 94850 USA

Geoffrey W. Diehl and Alik S. Widge are with the Translational Neuroengineering Laboratory, Department of Psychiatry and Behavioral Sciences, University of Minnesota, Minneapolis, MN 55455 USA.

Volkan Cevher is with the Institute of Electrical and Micro Engineering, École Polytechnique Fédérale de Lausanne, 1015 Lausanne, Switzerland.

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intrinsic phase of pathological oscillations, phase-locked DBS can enhance neural entrainment and synaptic plasticity, thus improving efficacy, reducing adverse effects, and sustaining long-term therapeutic benefits [2], [18], [19].

From a neuroscientific perspective, network-level synchrony biomarkers such as phase locking value (PLV) and phase-amplitude coupling (PAC) have been proposed as indicators of psychiatric and affective states [20], [21], [22], [23], [24], [25]. Disruptions in cross-regional synchronization of neural oscillations are associated with pathological states in various neurological and psychiatric disorders [6], [19], [21], [22], [23], [25]. Therefore, combining phase-locking events with connectivity measures (multivariate biomarkers), along with spectral energy (SE) and phase ($\phi_{\mathrm{LFP}}$) measures (univariate biomarkers) [26], [27], may provide a powerful framework for multi-mode stimulation control in the treatment of diverse psychiatric disorders. In addition, integrating inherently interpretable classifiers, such as neural additive models (NAMs), is a promising approach for dynamic DBS regulation, as they provide feature-wise contribution transparency for psychiatric therapy [6], [20], [24], [28].

Accurate and energy-efficient computation of these biomarkers, along with low-latency classification of psychiatric brain states, is essential for effective closed-loop neuromodulation [29]. However, current implantable systems face two major challenges. First, on-chip feature extraction (FE) engines either consume excessive power or rely on coarse approximations that compromise accuracy. For instance, CORDIC-based architectures [29], [30] compute instantaneous phase, PLV, and PAC, but at the cost of high energy consumption. In contrast, recent work has explored approximation-based techniques to reduce power consumption. However, these approaches often compromise the accuracy of extracted biomarkers [9], [19], [31]. Second, most closed-loop DBS systems targeting psychiatric disorders still rely on threshold-based control of extracted biomarkers to infer symptom states [9]. Although these methods are computationally simple and energy-efficient, they cannot model the complex, nonlinear dynamics of psychiatric disorders, leading to suboptimal therapeutic outcomes [6]. Alternatively, machine learning (ML)-based decoders can learn complex mappings from synchrony-based features such as PLV and PAC [6], [20], [22], [23] to latent psychiatric brain states, enabling more accurate and adaptive stimulation control.

To address these limitations, we present a 32-channel brain–computer interface (BCI) system-on-chip (SoC) that implements the envisioned architecture shown in Fig. 1 [6] and incorporates the following key components.

1) A low-noise, 32-channel analog front-end (AFE) with a programmable switch matrix for flexible neural signal acquisition.
2) A custom logarithmic arithmetic unit (LAU) for efficient phase-based and spectral FE with substantial power and area savings.
3) A NAM classifier with a test-time template adjuster (T3A) update mechanism that provides low-complexity, interpretable, and adaptive decoding of brain states.

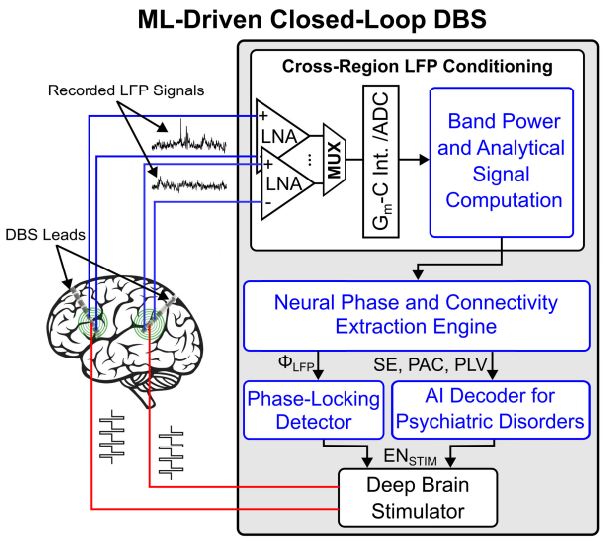



Fig. 1. Conceptual block diagram of an ML-driven closed-loop DBS system with phase-locked stimulation capability for psychiatric disorders [6].

4) A programmable, multi-mode stimulation controller supporting various feature-based, phase-locked, and ML-driven neuromodulation modes.

Extending upon our prior work [32], the proposed architecture enables real-time, biomarker-driven neuromodulation with improved accuracy, energy efficiency, and adaptability, paving the way for next-generation DBS systems tailored to psychiatric applications. To the best of our knowledge, this is the first SoC to support both ML-driven and phase-locked neuromodulation within a low-power, programmable platform.

Our paper is structured as follows. Section II presents an overview of the SoC architecture, followed by detailed descriptions of the hardware implementations of the AFE, feature extractor, NAM classifier, and stimulation control modules. Section III reports the SoC's experimental results, including electrical, in vitro, and in vivo validations. Finally, Section IV concludes this article.

## II. Psychiatric BCI SoC Architecture

Fig. 2 illustrates the top-level architecture of the proposed SoC. Multiple leads, comprising up to 64 individual electrodes, are strategically placed across multiple brain networks to identify optimal sites for connectivity extraction and subsequent stimulation. The switch matrix preceding the low-noise amplifiers (LNAs) allows flexible configuration of the recording setup, enabling selection between bipolar and monopolar electrode combinations. The 32-channel AFE comprises two modules, each integrating 16 LNAs that share a $G_m$–$C$ integrator and a 10-bit SAR ADC. The digitized neural signals from the AFE are processed by two sets of LAU-based FEs that compute SE, phase, amplitude envelope, and connectivity features. The proposed LAUs are custom-designed arithmetic units that enable low-power computation of multiplication, division, exponential, and root operations with high accuracy. The extracted features are then fed into an adaptive NAM classifier optimized for hardware implementation via knowledge

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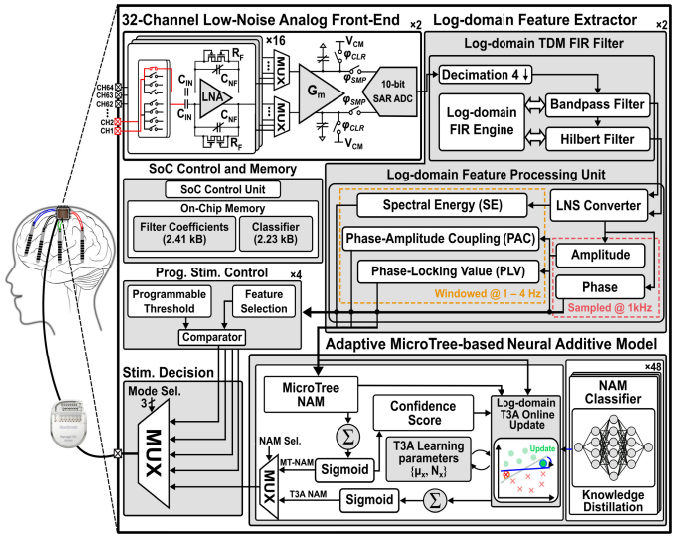



Fig. 2. System architecture of the proposed 32-channel psychiatric BCI SoC, introducing logarithmic FE and adaptive NAM-based psychiatric state detection.

distillation and implemented using low-complexity comparator trees, referred to as micro trees (MTs).

In addition, the extracted features are sent to multiple programmable thresholding blocks. Each block supports configurable Boolean logic (AND/OR) to combine per-sample phase triggers with window-level thresholded biomarkers, enabling flexible therapeutic policies (e.g., phase-locked stimulation enabled only when PLV exceeds a predefined threshold). The thresholded feature outputs together with the classifier output are then provided to a stimulation decision block that multiplexes among these inputs to generate the final DBS trigger.

### A. Low-Noise AFE

A low-noise AFE for multi-channel neural recording must satisfy key requirements such as low input-referred noise (IRN), effective electrode DC offset (EDO) rejection, high input impedance, and flexible electrode configurability. Fig. 3(a) illustrates the AFE architecture and the flexible electrode interface. The AFE comprises 32 channels, organized into two modules, each containing 16 LNAs, a charge-sampling $G_m$–$C$ integrator, and a 10-bit SAR ADC.

As shown in Fig. 3(b), a programmable switch matrix is placed before the LNAs to flexibly route electrode signals to the amplifier inputs. Each group of four AFE channels can interface with eight electrodes on a DBS lead, allowing flexible selection of input pairs for differential recording. This supports various sensing modes, including both monopolar and bipolar configurations. In the widely used monopolar configuration, multiple electrodes are referenced to a common electrode [e.g., electrode 1 in Fig. 3(b)], enabling the capture of widespread local field potentials (LFPs) but with increased susceptibility to far-field interference. In contrast, the bipolar configuration employs closely spaced electrode pairs to suppress distant neural interference and enhance spatial specificity. This flexibility is essential for accurate neural source localization,

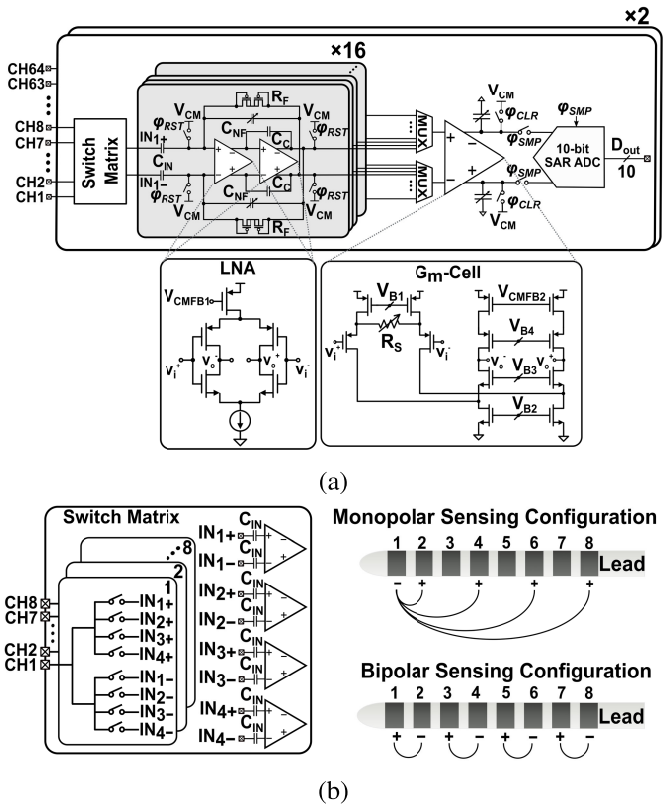



Fig. 3. (a) Circuit implementation of the 32-channel low-noise AFE and (b) switch matrix implementation and flexible electrode configuration.

hypersynchrony mapping, and analysis of DBS-induced therapeutic effects [33]. In particular, monopolar support is advantageous for implementation simplicity and scalability, whereas direct bipolar sensing retains benefits in analog-domain artifact and common-mode rejection. This is particularly important in closed-loop neuromodulation where large artifacts are common, such as stimulation-induced interference from the device itself [34], physiological or motion-related artifacts [35], and volume-conducted signals from nearby tissue that are not relevant to the target recording [36]. For these reasons, bipolar sensing is often preferred in practical closed-loop applications.

The input switch matrix is sized by trading off ON-resistance against parasitic loading at the input nodes. Although larger switches reduce the ON-resistance, they also increase parasitic loading, which can degrade the input impedance and common-mode rejection ratio (CMRR). The chosen sizing (1.6 $\mu$m/280 nm) achieves sufficiently low ON-resistance with only a modest impact on the input interface. In the intended closed-loop operation, the input switch matrix is not directly exposed to large stimulation-induced common-mode excursions. System-level protection is provided by an off-chip high-voltage switch, and blanking is applied during stimulation so that the recording path is disabled during large-artifact intervals.

Recent AFE architectures often incorporate some combination of chopper stabilization, DC servo loops (DSLs), and positive-feedback structures to mitigate low-frequency noise, suppress EDOs, and enhance input impedance [37],

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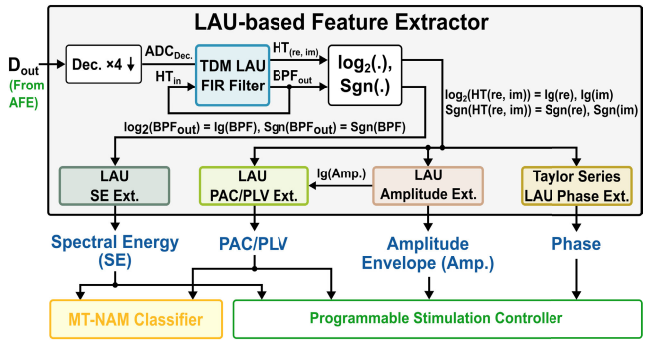
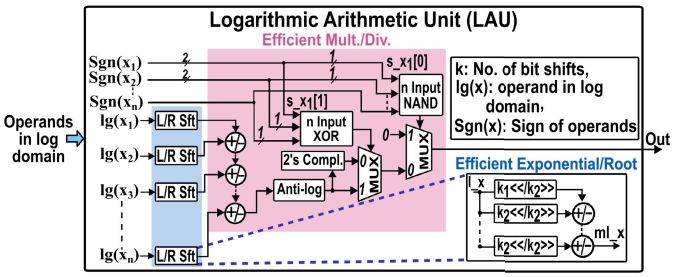



Fig. 4. Architecture of the proposed LAU-based feature extractor.




(a)

**LAU vs. Conventional Multiplier Hardware**

| Operation | Area (μm²) | Power (nW) |
|---|---|---|
| Conventional 10-bit Multiplier | 765.36 | 9.06 |
| 10-bit LAU Multiplier | 364.68 | 3.56 |
| 10-bit LAU Divider | 395.68 | 3.78 |
| 10-bit LAU Square | 78.50 | 0.92 |
| 10-bit LAU Cube | 398.00 | 3.98 |
| 10-bit LAU Root | 76.50 | 0.88 |

LAU operations assume operands in the log domain.
Log conversion overhead: 145.44 μm² and 1.02 nW.

(b)

Fig. 5. (a) Architecture of the proposed reconfigurable multi-input LAU and (b) hardware comparison of the proposed LAU with a conventional 10-bit fixed-point multiplier.

[38], [39]. In contrast, our design operates without chopper stabilization, as sufficient phase-locking accuracy can be achieved with an IRN level below 10 $\mu V_{rms}$ [19]. Each LNA adopts an AC-coupled closed-loop architecture with 10-pF metal–insulator–metal (MIM) input capacitors ($C_{IN}$), and a current-reuse topology with large input transistors for efficient suppression of 1/f noise. The 60-pF compensation capacitor ($C_c$) is implemented using 2.5-V thick-oxide NMOS capacitors, providing substantial area savings with negligible impact on linearity and IRN. Drift is further suppressed by the AC-coupled input stage, which removes EDO and low-frequency baseline variations before amplification. In addition, bipolar sensing attenuates shared DC fluctuations and common-mode interference at the input, improving robustness against baseline drift and artifacts. The measured high-pass corner of 0.22 Hz ensures that residual drift lies outside the signal band of interest and does not affect FE. Upon power-up, the virtual ground and output nodes are briefly reset to the common-mode voltage ($V_{CM}$) to ensure rapid settling of the LNAs through pseudo-resistors.

The LNA outputs are multiplexed to a shared $G_m$–$C$ integrator that performs charge-domain integration. This integrator provides a sinc-shaped frequency response for alias suppression and features programmable gain control through a 3-bit resistor bank ($R_S$) and a 4-bit capacitor bank ($C_{INT}$). The integrated signals are then digitized by a 10-bit asynchronous SAR ADC operating at an aggregate sampling rate of 64 kS/s (4 kS/s per channel). The ADC employs a binary-weighted CDAC with monotonic switching and bootstrapped sampling switches, achieving a conversion time of 312.5 ns. Asynchronous SAR control contributes to reduced power consumption and area-efficient design [40].

### *B. Efficient Logarithmic-Domain Biomarker Extraction*

Fig. 4 illustrates the proposed LAU-based FE module, designed for real-time, low-power neural biomarker computation in adaptive DBS systems. Decimated neural signals from the AFE are processed by a time-division multiplexed (TDM) finite impulse response (FIR) filter, implemented using the LAU to perform both bandpass filtering (BPF) and Hilbert transform (HT) through hardware reuse. The resulting analytic signal is then converted to the log number system (LNS) [$\log_2(\cdot)$, $\mathrm{sgn}(\cdot)$] to facilitate log-domain feature computation. The SoC generates two bandpass-filtered signals: one used for SE and a second used for connectivity features (PAC or PLV). These two bands are independently configurable and can be selected from a set of eight predefined LFP bands spanning 1–500 Hz (e.g., delta, theta, alpha, beta, gamma, and higher-frequency bands). This allows flexible selection of different frequency bands per channel.

The LAU-based FE module computes: 1) SE from the BPF output; 2) instantaneous amplitude; 3) instantaneous phase; and 4) PAC or PLV. As shown in Fig. 4, signals are converted to the logarithmic domain once after filtering, and the resulting representation is reused across multiple operations (e.g., phase, amplitude, PAC, and PLV), thereby reducing redundant conversions. In addition, logarithmic conversions are implemented using Mitchell's scheme [41], [42], enabling low-cost log/antilog operations. The minimized number of conversions, together with the low-complexity log/antilog implementation, results in a significant reduction in overall computational complexity. The extracted features are then routed to a programmable classifier and a multi-mode thresholding circuit.

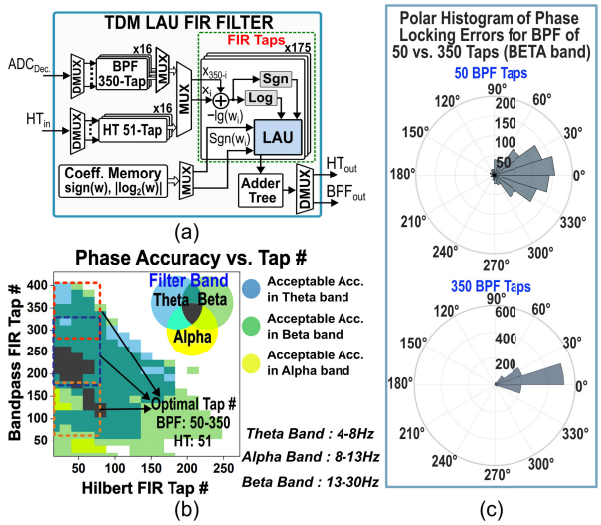
*1) Logarithmic Arithmetic Unit (LAU):* The FE stage involves arithmetic operations such as multiplication, division, exponentiation, and trigonometric functions, all of which are computationally intensive in conventional digital implementations. Prior work has employed approximation-based techniques to reduce power [9], [19], [31], often at the expense of accuracy, which is an undesirable tradeoff in BCI applications where precise biomarker computation is critical for reliable closed-loop performance.

To address this challenge, we propose a reconfigurable LAU that can be configured according to the required function

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Floating Point MATLAB vs. LAU FIR Filter (350-tap)

| BPF Band | Correlation (%) | NRMSE (%) | SER (dB) | RMS Error (%) | Energy Error (%) | Gain Error (%) |
|---|---|---|---|---|---|---|
| Theta | 99.99 | 0.0967 | 37.71 | 0.0766 | 0.1533 | 0.0850 |
| Alpha | 99.98 | 0.1579 | 31.99 | 0.0254 | 0.0508 | 0.0570 |
| Beta | 99.94 | 0.2109 | 28.50 | 0.2739 | 0.5485 | 0.3375 |

(d)

Fig. 6. (a) Architecture of the proposed TDM LAU-based FIR filter, (b) phase accuracy versus tap count with software-defined accuracy thresholds [43], (c) polar histogram of phase-locking errors for 50- and 350-tap FIR filters, and (d) performance comparison between floating-point MATLAB and LAU-based 350-tap FIR filter implementations across theta, alpha, and beta bands, showing high correlation and low error metrics.

to perform fixed-point arithmetic operations directly in the logarithmic domain, as shown in Fig. 5(a). This allows complex mathematical computations to be realized with simple digital hardware. Specifically, the proposed LAU supports multiplication, division, exponentiation, and root extraction primarily through low-complexity shifters and adders. Logarithmic conversion, $\log_2(\cdot)$, is computed with a leading-one detector and barrel shifter, while anti-$\log_2(\cdot)$ is realized with a barrel shifter. The signum function, sgn(·), is implemented using simple comparators.

The area and power consumption of key arithmetic operations (multiplication, division, root, and exponentiation) implemented using the proposed LAU are compared against conventional 10-bit fixed-point implementations under the same precision constraints and are shown in Fig. 5(b). The results show that the LAU significantly reduces hardware complexity, leading to substantial savings in both area and power. By reformulating mathematical functions to operate in the log domain, the reconfigurable LAU enables efficient realization of diverse operations while achieving a favorable energy–accuracy tradeoff, making it well suited for low-power BCI SoC applications.

*2) Bandpass Filter Tap Count Selection:* The initial stage of FE involves multiple filtering operations. A BPF is first applied for SE extraction, followed by another BPF and an HT to compute the analytic signals, providing real and imaginary components to derive phase, amplitude envelope, and connectivity features. All filtering operations are implemented using a TDM FIR filter that supports up to 350 taps, as shown in Fig. 6(a). To reduce computational complexity, the FIR employs multiplication through the proposed LAU. The filter adopts a symmetric structure to reduce memory requirements and the number of multiplications, while supporting configurable tap counts from 50 to 350. Prior studies [43], [44] [see Fig. 6(b)] and our experimental results [see Fig. 6(c)] indicate that increasing the tap count improves the spectral isolation of target oscillations and enhances the accuracy of extracted biomarkers [43]. Particularly, as shown in Fig. 6(b), the SoC's phase extraction performance was evaluated across different frequency bands. Each band was tested with three independently generated signals containing both in-band and out-of-band oscillatory components. The shaded regions indicate where the SoC's performance exceeded the acceptable accuracy threshold defined by the real-time Toolkit for Oscillatory Real-Time Tracking and Estimation (TORTE) algorithm [45]. Similar evaluations were performed for the PAC and PLV metrics [43]. Although increasing the number of FIR taps improves frequency selectivity, it does not necessarily improve phase accuracy for neural signals. The optimal tap count is determined by both the spectral characteristics of the oscillation and the requirements of phase estimation, rather than by filter sharpness alone. Neural oscillations are inherently nonstationary and occupy a finite bandwidth around their center frequency [43], rather than being perfectly narrowband. Therefore, the system supports configurable tap counts to enable an appropriate tradeoff between spectral isolation and phase accuracy across different frequency bands and extracted features.

Fig. 6(c) presents polar histograms of phase-locking errors computed from pre-recorded LFP in a Long-Evans rat. In both 50- and 350-tap configurations, stimulation was triggered at the 180° phase of the beta band (13–30 Hz). The corresponding FIR group delays were 49.5 and 199.5 ms, resulting in phase pre-shifts of 107° and 430°, respectively, at a 21-Hz center frequency. The phase-locking error, defined as the deviation from the target phase at the stimulation instant, was evaluated against a ground-truth reference generated using a zero-phase second-order Butterworth filter and an ideal HT in MATLAB [43]. The 350-tap configuration yields a tighter clustering around 0°, indicating lower phase error and improved timing precision for beta-band alignment compared to the 50-tap configuration. These results highlight the benefit of higher tap counts in the BPF. The proposed programmable FIR architecture supports eight tap configurations (50, 100, 150, 200, 250, 300, 320, and 350), and the filters are clock-gated based on the selected tap configuration to reduce power consumption. Fig. 6(d) compares the performance of the 350-tap LAU-based FIR BPF with that of a floating-point multiplier-based FIR for the theta, alpha, and beta bands, using identical ADC inputs from 60 s of prerecorded neural data. The results show that the signal-to-error ratio (SER) ranges from 28.50 to 37.71 dB, corresponding to an error below 0.13% in the signal power, even after accumulation over 350 taps. In addition, the normalized RMSE remains below

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0.21%, the band-energy error below 0.55%, and the gain error below 0.34%. The Pearson correlation between the LAU-based and ideal outputs exceeds 0.999 across all bands, confirming negligible degradation in filter accuracy.

*3) TLP Extractor:* The arctangent of the analytic signal is widely used for real-time phase extraction in neural interfaces [9], [46]. However, conventional implementations such as CORDIC processors achieve high accuracy at the expense of substantial power and area, making them unsuitable for implantable devices. A light phase extractor (LPE) was recently proposed, which approximates the phase using a first-order Lagrange interpolation to reduce hardware complexity while introducing negligible accuracy loss [19]. Nevertheless, the LPE still requires multipliers and lookup tables (LUTs). To further reduce power and area, we propose a Taylor-series-based LAU phase (TLP) extractor that enables on-chip phase computation with even lower hardware overhead and negligible accuracy loss. For a complex signal $z = a + ib$, the normalized arctangent can be expanded as

$$\frac{\tan^{-1}\left(\frac{b}{a}\right)}{\pi} = \begin{cases} -\frac{1}{2} - \left(\frac{a}{\pi b} - \frac{a^3}{3\pi b^3} + \frac{a^5}{5\pi b^5} - \cdots\right), & \text{if } \frac{b}{a} < -1 \\ \frac{1}{2} - \left(\frac{a}{\pi b} - \frac{a^3}{3\pi b^3} + \frac{a^5}{5\pi b^5} - \cdots\right), & \text{if } \frac{b}{a} > 1 \\ \frac{b}{\pi a} - \frac{b^3}{3\pi a^3} + \frac{b^5}{5\pi a^5} - \cdots, & \text{otherwise.} \end{cases} \quad (1)$$

TLP employs a truncated Taylor expansion (1), where retaining the first three terms provides sufficient accuracy for a 10-bit fixed-point implementation, as verified through our MATLAB simulations. Although the Taylor series inherently involves cubic terms, multiplications, and divisions, reformulating the computation in the logarithmic domain using the LAU substantially reduces hardware complexity and energy consumption. The reformulated arctangent for the analytic signal can then be expressed as

$$\frac{\tan^{-1}\left(\frac{\text{im}}{\text{re}}\right)}{\pi} \approx \begin{cases} -\frac{1}{2} - \left(s\,2^{-x-K_1} - s\,2^{-3x-K_2}\right), & \frac{\text{im}}{\text{re}} < -1 \\ \frac{1}{2} - \left(s\,2^{-x-K_1} - s\,2^{-3x-K_2}\right), & \frac{\text{im}}{\text{re}} > 1 \\ s\,2^{x-K_1} - s\,2^{3x-K_2}, & \text{otherwise} \end{cases} \quad (2)$$

where

$$x = \log_2\left|\frac{\text{im}}{\text{re}}\right|, \quad s = \text{Sgn(im)} \cdot \text{Sgn(re)}$$
$$K_1 = \log_2(\pi), \quad K_2 = K_1 + \log_2(3).$$

The final phase is assigned to the correct quadrant based on the signs of re and im.

As illustrated in Fig. 7(a), the proposed TLP module enables an efficient implementation of (2). Hardware synthesis results in Fig. 7(b) indicate that the TLP achieves 73.5% and 44.7% reductions in area, and 67.4% and 53.0% lower power consumption compared to unrolled CORDIC and LPE implementations [19], respectively, under a consistent 10-bit resolution. These improvements make the TLP well-suited for integration in low-power BCI designs.

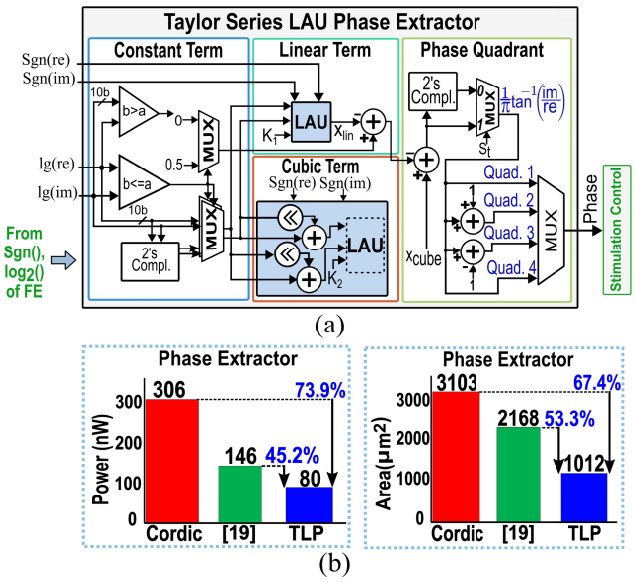



Fig. 7. (a) Architecture of the proposed TLP and (b) hardware comparison with state-of-the-art designs for 10-bit implementations.

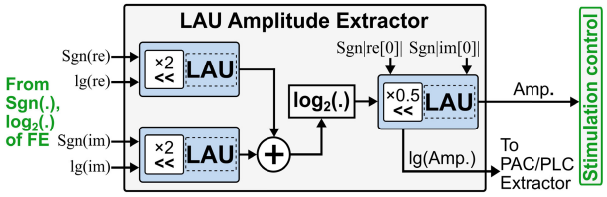



Fig. 8. Architecture of the proposed LAU-based amplitude envelope extractor.

*4) Amplitude Envelope (Amp):* The $l_2$-norm (amplitude) of a complex number $z = a + ib$ is given by

$$Amp = \sqrt{a^2 + b^2}.$$

However, the computation of the $l_2$-norm involves both square and square-root operations, which are power- and area-intensive in hardware. Consequently, some prior works approximate amplitude using the $l_\infty$-norm, trading accuracy for efficiency [9], [19], [31]. In contrast, expressing the $l_2$-norm computation in the LNS domain allows the square and square-root operations to be implemented directly by the LAU, as shown in Fig. 8. Taking the base-2 logarithm of both sides of the analytic signal yields

$$\begin{aligned} \log_2(\text{Amp}) &= 0.5\log_2(\text{re}^2 + \text{im}^2) \\ &= 0.5\log_2\left(|\text{Sgn(re)}|\,2^{2\log_2|\text{re}|}\right. \\ &\quad \left. + |\text{Sgn(im)}|\,2^{2\log_2|\text{im}|}\right) \end{aligned} \quad (3)$$

where $\log_2(\text{Amp})$ is directly utilized by the PAC/PLV extractor, as illustrated in Fig. 4. The final amplitude envelope is then obtained by

$$\text{Amp} = |\text{Sgn(re)}| \vee |\text{Sgn(im)}| \cdot 2^{\log_2(Amp)}. \quad (4)$$

*5) Spectral Energy (SE):* The SE of a band-passed signal $X$ with $N$ samples is defined as

$$\text{SE} = \frac{1}{N}\sum_{i=1}^{N} X(i)^2. \quad (5)$$

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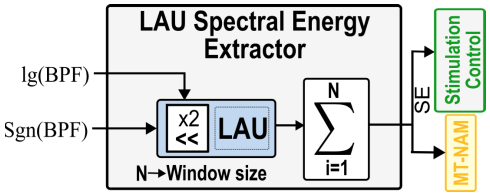



Fig. 9. Architecture of the proposed LAU-based spectral energy extractor.

To enable low-complexity hardware implementation using LNS, each squared term can be reformulated as

$$\mathrm{SE} = \frac{1}{N}\sum_{i=1}^{N} 2^{2\log_2|X(i)|}. \tag{6}$$

This formulation eliminates the need for multipliers and enables computation using LAU, as shown in Fig. 9.

*6) PLV and PAC:* The PLV quantifies phase synchronization across channels, while PAC measures the degree to which the phase of a low-frequency oscillation modulates the amplitude of high-frequency activity. Conventional PLV and PAC computations rely on analytic signals, followed by operations involving instantaneous phase, amplitude, and trigonometric functions [29], [38]. The mathematical definitions of PLV and PAC are given by

$$\mathrm{PLV} = \frac{1}{N}\sqrt{\left(\sum_{t=1}^{N}\sin(\Delta\theta_t)\right)^2 + \left(\sum_{t=1}^{N}\cos(\Delta\theta_t)\right)^2} \tag{7}$$

$$\mathrm{PAC} = \frac{1}{N}\sqrt{\left(\sum_{t=1}^{N}A_t\sin(\theta_t)\right)^2 + \left(\sum_{t=1}^{N}A_t\cos(\theta_t)\right)^2}. \tag{8}$$

Here, $\Delta\theta_t$ denotes the cross-channel phase difference, whereas $\theta_t$ and $A_t$ represent the modulating phase and amplitude envelope for PAC, respectively, derived from the selected frequency bands. To reduce computational complexity and enable hardware-efficient implementation, we reformulate (7) and (8) to eliminate explicit phase and trigonometric computations. The proposed approach computes PLV and PAC directly from the real and imaginary components of the analytic signal (re and im), their delayed counterparts ($\mathrm{re}_D$ and $\mathrm{im}_D$), and the corresponding Amp ($A$ and $A_D$). The delay $D$ corresponds to one sampling clock, as shown below

$$\begin{aligned} A_D\sin(\theta) &= A_D\frac{\mathrm{im}}{A} \\ A_D\cos(\theta) &= A_D\frac{\mathrm{re}}{A} \end{aligned} \tag{9}$$

$$\begin{aligned} \sin(\theta-\theta_D) &= \frac{\mathrm{im}\cdot\mathrm{re}_D - \mathrm{re}\cdot\mathrm{im}_D}{A\cdot A_D} \\ \cos(\theta-\theta_D) &= \frac{\mathrm{re}\cdot\mathrm{re}_D + \mathrm{im}\cdot\mathrm{im}_D}{A\cdot A_D}. \end{aligned} \tag{10}$$

To further reduce computational cost, the operations are reformulated using the LNS

$$\begin{aligned} A_D\sin\theta &= \mathrm{Sgn}(\mathrm{im})\ 2^{\log_2 A_D+\log_2|\mathrm{im}|-\log_2 A} \\ A_D\cos\theta &= \mathrm{Sgn}(\mathrm{re})\ 2^{\log_2 A_D+\log_2|\mathrm{re}|-\log_2 A} \\ \sin(\theta-\theta_D) &= s_1\ 2^{\log_2|\mathrm{im}|+\log_2|\mathrm{re}_D|-\ell_A} \end{aligned} \tag{11}$$

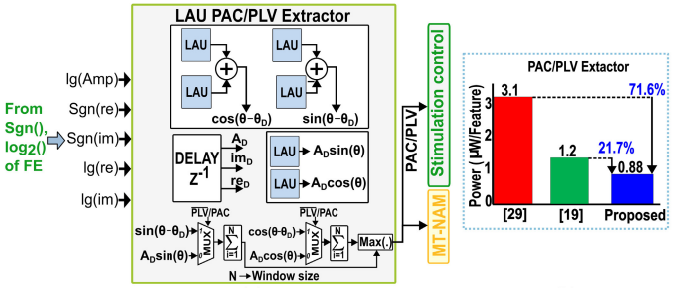



Fig. 10. (a) Proposed LAU-based PAC/PLV extractor and (b) hardware comparison with state-of-the-art implementations.

$$\begin{aligned} \cos(\theta-\theta_D) = {} & -s_2\ 2^{\log_2|\mathrm{re}|+\log_2|\mathrm{im}_D|-\ell_A} \\ & s_3\ 2^{\log_2|\mathrm{re}|+\log_2|\mathrm{re}_D|-\ell_A} \\ & + s_4\ 2^{\log_2|\mathrm{im}|+\log_2|\mathrm{im}_D|-\ell_A} \end{aligned} \tag{12}$$

where $s_1 = \mathrm{Sgn}(\mathrm{im})\cdot\mathrm{Sgn}(\mathrm{re}_D)$, $s_2 = \mathrm{Sgn}(\mathrm{re})\cdot\mathrm{Sgn}(\mathrm{im}_D)$, $s_3 = \mathrm{Sgn}(\mathrm{re})\cdot\mathrm{Sgn}(\mathrm{re}_D)$, $s_4 = \mathrm{Sgn}(\mathrm{im})\cdot\mathrm{Sgn}(\mathrm{im}_D)$, and $\ell_A = \log_2 A + \log_2 A_D$. This formulation supports PAC and PLV computations that require phase and amplitude information from distinct frequency bands, as well as the phase difference across those bands.

The proposed reformulation eliminates explicit phase extraction and costly trigonometric LUTs, significantly reducing computational and hardware overhead to enable efficient implementation of (9) and (10). For the aggregation stage in (7) and (8), the magnitude is approximated using the $\ell_\infty$-norm [i.e., $\max(|X|,|Y|)$], since this operation is applied to accumulated terms and is less sensitive to small relative errors as shown in [9], [19], and [31]. The proposed feature extractor, as shown in Fig. 10(a), can simultaneously compute eight PAC or PLV features, with each channel pair configurable to output either metric. The SoC includes two such FE blocks operating in parallel (see Fig. 2) resulting in a total of 16 PAC or PLV features. The total power dissipation is 0.88 $\mu$W per extractor, using the 50-tap FIR filter. A comparison of the per-feature power consumption is shown in Fig. 10(b). The proposed LAU-based implementation occupies 7400 $\mu\mathrm{m}^2$ (excluding the FIR block), achieving a 25.4% area reduction compared to the 9925 $\mu\mathrm{m}^2$ reported in the state of the art [9].

### C. Statistical Feature Selection

The biomarkers generated by the LAU-based FE module are selected through an offline, hardware-informed feature selection process. This step is essential for identifying a subset of maximally informative neural features that enable accurate classification by the MicroTree NAM (MT-NAM) model while strictly adhering to the computational and memory constraints of the SoC.

Initially, the extracted biomarkers (32 SE and 16 PLV/PAC) are quantitatively evaluated and ranked using analysis of variance (ANOVA) F-scores. ANOVA is selected as a feature selection method due to its simplicity and model-agnostic filter-based formulation, which decouples feature selection from the downstream classifier and avoids the overfitting risks

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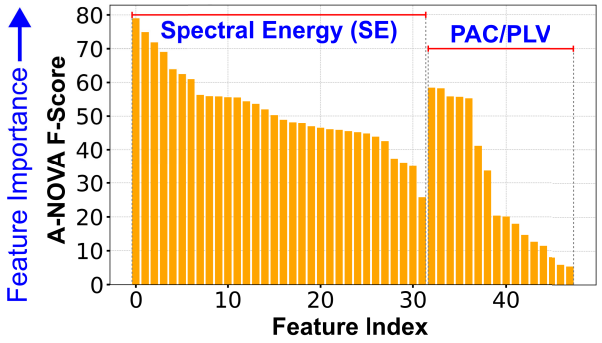



Fig. 11. Relative importance of features ranked by ANOVA F-scores and selection of the 48 most relevant features for a patient performing the ECR task.

associated with wrapper-based approaches. This statistical metric assesses the significance and discriminative ability of each biomarker with respect to the class labels as follows:

$$F(x_t, y_t) = \frac{\sum_{j=1}^{C} n_j(\bar{x}_j - \bar{x})^2/(C-1)}{\sum_{j=1}^{C}\sum_{t:y_t=j}(x_t - \bar{x}_j)^2 \Big/ \left(\sum_{j=1}^{C} n_j - C\right)}$$

$$\bar{x}_j = \frac{1}{n_j}\sum_{t:y_t=j} x_t, \quad \bar{x} = \frac{1}{\left(\sum_{j=1}^{C} n_j\right)}\sum_{t=1}^{T} x_t \tag{13}$$

where $(x_t, y_t)$ denotes an observation pair comprising a biomarker and its class label at time $t$, $\bar{x}_j$ is the mean value over samples from class $j$, $\bar{x}$ represents the global mean across all samples, $C$ is the total number of classes, $T$ is the total number of samples, and $n_j$ denotes the number of samples in class $j$. The biomarkers from all 32 neural recording channels are globally ranked based on their calculated ANOVA F-scores. To satisfy hardware constraints, the feature selection process allocates fixed quotas for each biomarker type, selecting 32 SE features and 16 PLV/PAC features. Candidate biomarkers are ranked using ANOVA F-scores and selected iteratively in descending order, subject to these predefined quotas. This allocation is consistent with the experimentally observed distribution of top-performing features (see Fig. 11), while preserving biomarker diversity for both classification and closed-loop operation, with no observable performance degradation compared to unconstrained ANOVA-based feature selection. The resulting compact subset balances classification performance with hardware efficiency.

### D. MicroTree-Based NAM

We employ a NAM-based classifier in our SoC, as it effectively captures complex non-linear relationships among extracted biomarkers while preserving interpretability [47], [48]. Such transparency is particularly critical in psychiatric and closed-loop neurostimulation applications, where opaque closed-box models are unsuitable for clinical decision-making. NAM operates by constructing a linear combination of feature-specific neural subnetworks, each learning a distinct, non-linear function for an individual input feature. These subnetworks are jointly trained using backpropagation with a binary cross-entropy loss. The contribution of each feature

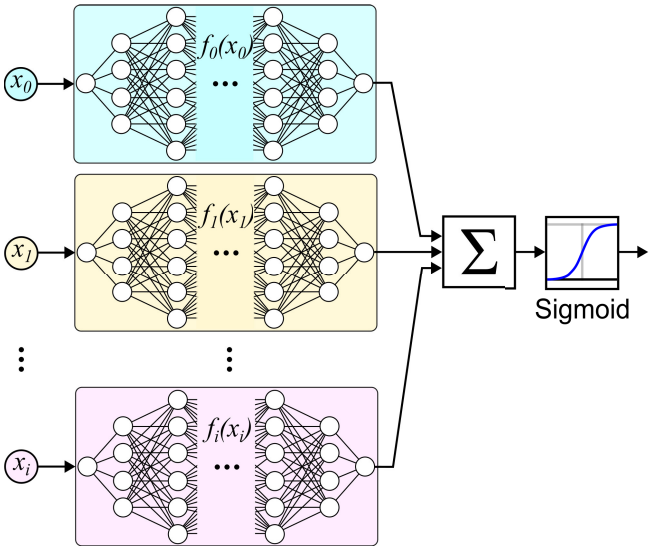



Fig. 12. NAM architecture. Each input feature is processed independently by a dedicated subnetwork, and the resulting outputs are accumulated to form the final prediction. This feature-specific structure enables the NAM to maintain interpretability of its learned representations.

can be interpreted by visualizing its learned shape function $f_i(x_i)$ as a function of the input $x_i$. During inference, each input feature is processed by its corresponding subnetwork, and the final prediction for time window $t$, $y_t$, is obtained by aggregating the outputs of all subnetworks, as illustrated in Fig. 12 and defined in the following equation:

$$y_t = \sigma\left(\sum_{i=1}^{N} f_i(x_{i,t})\right) \tag{14}$$

where $y_t$ is the model prediction, $x_{i,t}$ represents the $i$th input feature at time $t$ (e.g., band-specific SE from a given channel), $f_i(\cdot)$ is the non-linear function learned by the corresponding feature network, $N$ denotes the total number of extracted features, and $\sigma(\cdot)$ represents the sigmoid activation function used to generate the final probability output.

*1) Psychiatric Datasets:* NAM was extensively validated on the human electroencephalography (EEG) affective dataset for emotion analysis (DEAP) [49] across the arousal and valence dimensions, as well as on two intracranial EEG (iEEG) psychiatric datasets: the multi-source interference task (MSIT) and the emotional conflict resolution (ECR) task [50].

In the DEAP tasks, participants viewed emotionally engaging video clips and rated their emotional responses on a 1–9 scale for arousal (intensity) and valence (pleasantness). Higher arousal scores indicate more intense emotional reactions, while higher valence scores correspond to more positive emotions. For binary classification, the ratings were divided into two classes: scores of 1–5 were labeled as class 0 (low arousal/valence) and scores of 6–9 as class 1 (high arousal/valence). In the MSIT task, a patient's engagement in cognitive conflict resolution during decision-making was assessed. Participants were presented with three numbers (e.g., 2, 2, and 1) and instructed to indicate the identity of the distinct target number (e.g., 1) while ignoring the distractors (e.g., 2). Similarly, in the ECR task, participants viewed emotional facial expressions (e.g., happy) overlaid with congruent

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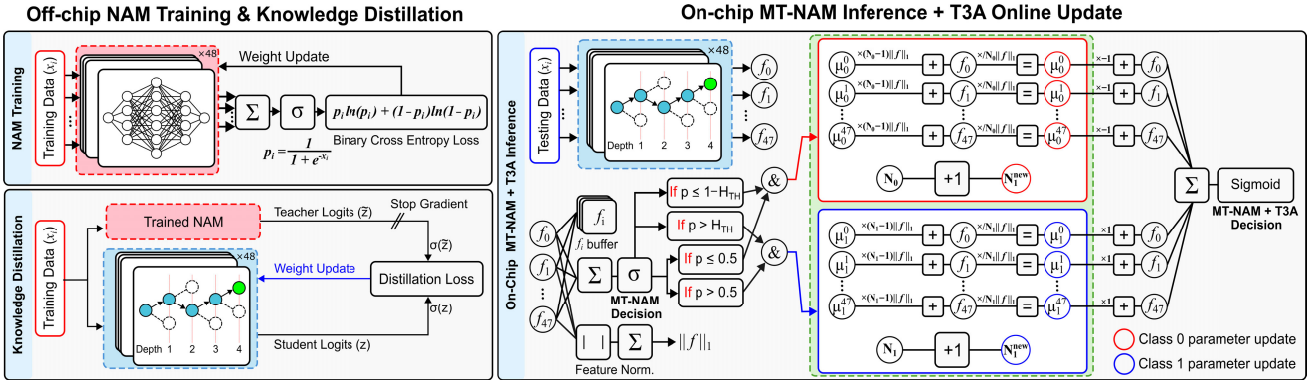



Fig. 13. Algorithmic flow of NAM with T3A update for offline training and on-chip inference. NAM is first trained offline using stochastic gradient descent with a binary cross-entropy loss. The trained NAM is then distilled into MT-NAM in a teacher–student framework by minimizing distillation loss, and deployed for on-chip inference with backpropagation-free T3A updates.

(e.g., happy) or incongruent (e.g., fear) text cues and were asked to identify the emotion while disregarding conflicting (e.g., fear) or nonconflicting (e.g., happy) texts as emotional cues. For both MSIT and ECR datasets, class labels correspond to task engagement (class 1) and resting periods (class 0). Prior work has shown that task engagement and conflict-resolution states can be more reproducible and informative for neuromodulation than directly classifying high-level psychiatric labels [8], [51], [52]. Therefore, detecting such intermediate states in MSIT and ECR provides appropriate and well-established validation scenarios for biomarker-driven closed-loop neuromodulation. For each dataset, the data is split into 80% training, 10% validation, and 10% testing sets on a per-patient basis, and patient-specific training and evaluation of the NAM were performed.

*2) MT-NAM Training:* The NAM architecture consists of feature-wise deep neural networks. Each network receives a scalar input corresponding to one feature, followed by three fully connected layers with 64, 64, and 32 neurons, respectively, and ends with a final output layer that generates a scalar output. The outputs from 48 such networks are then summed, and a sigmoid activation is applied to generate the final probability. However, these neural networks are computationally intensive for area- and power-constrained BCI SoCs due to the high cost of multiply-and-accumulate (MAC) operations. To address this challenge, we employ a knowledge distillation strategy, where the pre-trained NAM serves as a teacher model to train 48 compact decision trees by matching their output logits, achieving comparable accuracy with significantly improved inference efficiency. In this teacher–student framework, the decision trees approximate the output distribution of the neural-network teacher. Thus, we introduce the MT-NAM, in which each feature function $f_i(x_{i,t})$ of the original NAM is replaced by a shallow decision tree with a maximum depth of 4. The algorithmic flow of the NAM Training and MT-NAM Inference architecture is illustrated in Fig. 13.

To enable deployment of MT-NAM on a resource-constrained SoC, we retrained NAM using quantization-aware training (QAT). Unlike post-training quantization where a full-precision model is later converted to fixed-point at the risk of abrupt accuracy loss, QAT maintains robust performance

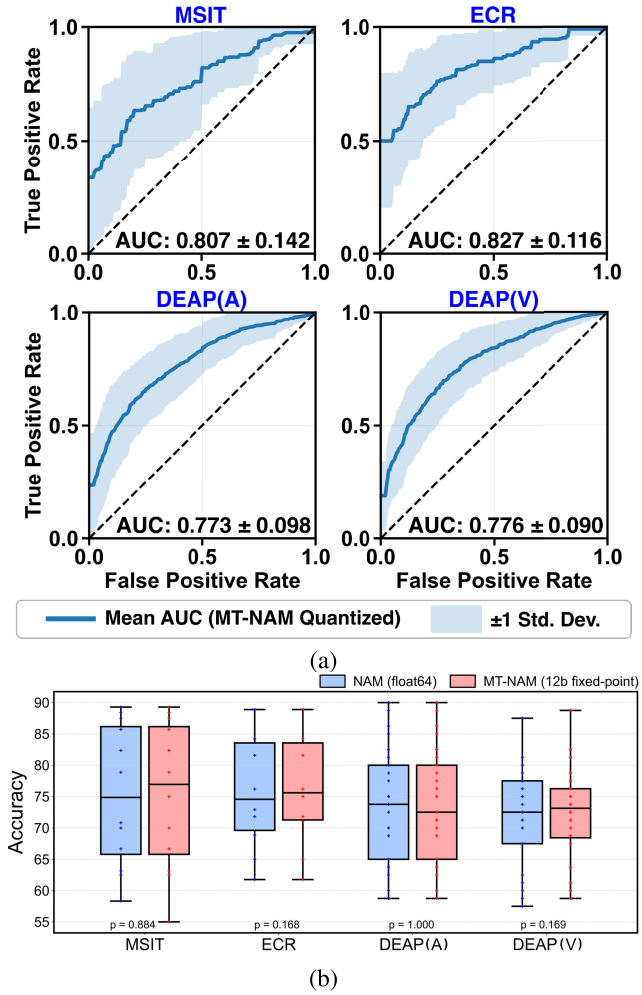



Fig. 14. (a) ROC characterization of the quantized MT-NAM classifier for MSIT, ECR, DEAP arousal [DEAP (A)] and DEAP valence [DEAP (V)]. The blue curve represents the mean ROC across patients, and the shaded region indicates the standard deviation. (b) Statistical comparison of double-precision floating-point NAM and 12-bit fixed-point MT-NAM accuracy using a paired-sample t-test.

by optimizing the model directly under 12-bit fixed-point constraints. The ROC curves in Fig. 14(a) further show that the quantized MT-NAM maintains robust, threshold-independent discrimination across datasets, as reflected by consistently high

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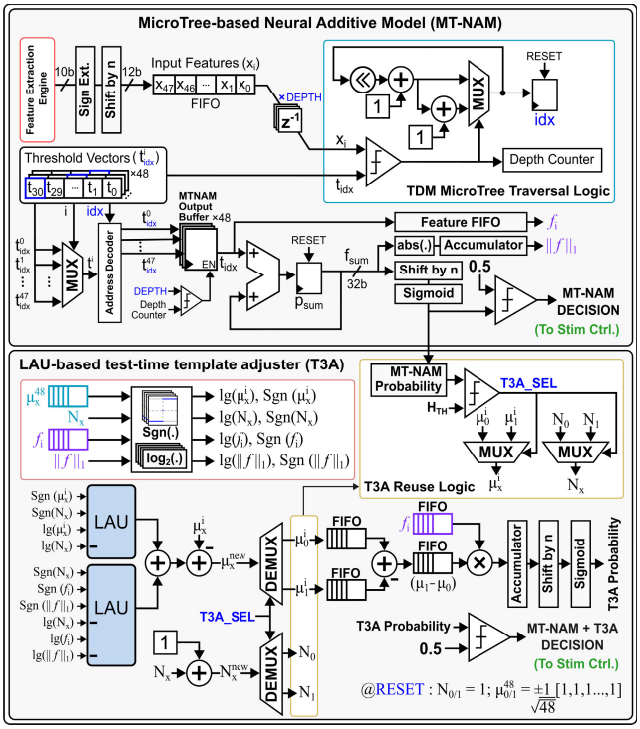



Fig. 15. TDM hardware implementation of the MicroTree-NAM with T3A update using the LAU.

AUC values. In addition, we employed early stopping based on validation loss to prevent overfitting. To assess any accuracy degradation introduced by knowledge distillation and QAT, we conducted a paired-sample t-test across all datasets, as shown in Fig. 14(b). The results show no statistically significant accuracy loss (i.e., $p \geq 0.05$), confirming the effectiveness of the combined distillation and QAT. For consistent evaluation across datasets, we used a fixed decision threshold of 0.5. In practice, however, this threshold remains tunable to enable patient-specific sensitivity-specificity operating points during calibration without modifying the underlying model.

*3) MT-NAM Architecture:* The core computational element of MT-NAM, as shown in Fig. 15, is the TDM MicroTree traversal module, which reuses a single comparator across all nodes of a time-division–multiplexed decision tree (configurable up to depth 4) to generate the output. The MT-NAM threshold vectors are stored in a level-order arrangement corresponding to each feature. A total of 31 thresholds per feature across 48 features (1488 values, 12 bits each, 2.24 kB) are held in the threshold memory and streamed to the TDM MicroTree. For each feature, its threshold vector is first loaded into the local register bank. At each node, the comparator evaluates whether the input feature value is greater than or less than the corresponding threshold, and the next node index is selected accordingly. If the current node index is $n \in \{0, 1, \ldots, 30\}$, $x_i$ is the input feature vector, and $t_{(n,xi)}$ its node threshold, then the next node index $n_{\text{next}}$ is computed as

$$n_{\text{next}} = \begin{cases} 2n+1, & \text{if } x_i < t_{(n,x_i)} \\ 2n+2, & \text{if } x_i \geq t_{(n,x_i)}. \end{cases} \tag{15}$$

When the depth counter reaches the configurable depth parameter $D$, the threshold stored at the final node index is forwarded to the accumulator. Furthermore, we prune the MicroTrees after knowledge distillation, which results in many trees having depths smaller than $D$. To correctly traverse these unbalanced trees, we assign a special marker value (the minimum in the normalized threshold range) to indicate the absence of a child node. When this marker is encountered, traversal terminates early, and the threshold from the last valid node index is output.

After each feature-specific tree traversal yields a scalar output for the current time window, these outputs are accumulated in a signed 32-bit register that continually updates the running partial sum. To accommodate variations in threshold ranges across patients, the accumulated value is scaled accordingly after each update. Finally, the combined output from all 48 features is passed to a 6-bit key sigmoid LUT, which maps the accumulated output to a probability between 0 and 1.

### *E. T3A Online Update*

T3A [53] serves as our backpropagation-free online adaptation strategy, enhancing classifier robustness under distributional shifts in incoming feature streams. T3A is particularly well suited for this application because it updates only a lightweight prototype layer rather than the full network, significantly reducing computational overhead. In addition, confidence-based thresholds are applied to regulate template updates and limit potential drift. Our method leverages high-confidence test-time predictions to iteratively refine class-specific templates (centroids), thereby adjusting the decision boundary during runtime.

*1) Centroid Initialization and Update Strategy:* At initialization, two centroids, $\mu_0$ and $\mu_1$, are assigned to represent the negative and positive classes, respectively. These centroids are configured as normalized vectors at SoC reset: $\mu_0 = (1/d)[1, 1, \ldots, 1]$ and $\mu_1 = -(1/d)[1, 1, \ldots, 1]$, where $d = 48$ is the dimension of the feature space. Each centroid is paired with an update counter, $N_0$ and $N_1$, both initialized to 1, which track the number of adaptation steps for their respective classes.

During inference, for each feature vector $f$, the classifier produces a confidence score from the sigmoid output. A configurable threshold ($H_{\text{TH}}$) ensures that only high-confidence predictions contribute to centroid adaptation. When a prediction's confidence exceeds this threshold, the corresponding centroid is updated using an online weighted moving average, as defined in the following equation:

$$\mu_x \leftarrow \underbrace{\left(1 - \frac{1}{N_x}\right)\mu_x}_{\text{Centroid memory}} + \underbrace{\frac{1}{N_x}\left(\frac{f}{\sum |f|}\right)}_{\text{Feature update}} \tag{16}$$

$$N_x \leftarrow N_x + 1 \tag{17}$$

where $x \in \{0, 1\}$ denotes the predicted class label. The first term in (16) preserves the contribution of the previous centroid $\mu_x$. As the update count $N_x$ increases, the weight $(1/N_x)$ assigned to the new feature decreases, providing stability and reducing sensitivity to later updates. The second term incorporates the feature update using the $l_1$-normalized feature vector, ensuring scale-invariant contributions so that no individual

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feature dimension disproportionately influences the centroid. As in the centroid-memory update, division by $N_x$ causes the impact of new samples to diminish with accumulated updates, promoting convergence of the running average.

*2) Decision Boundary Update:* The T3A update mechanism ensures that the centroids $\mu_0$ and $\mu_1$ gradually adapt toward the true data distribution over time as new feature vectors arrive at runtime, thereby adjusting the decision boundary to minimize misclassifications. Updating the centroids using only high-confidence predictions ensures that the refined templates more accurately represent the incoming feature distribution, leading to improved classification performance. The updated centroids are then used to generate the T3A prediction for subsequent samples by applying a sigmoid function to the computed similarity measure

$$y = \sigma\left((\mu_1 - \mu_0)^\top f\right) \tag{18}$$

where $y$ is the updated probability prediction. This continuous centroids refinement enables T3A to effectively mitigate domain shift, making it well-suited for online and adaptive inference scenarios.

*3) LAU-Based T3A Update:* We reduce the computational complexity of the T3A update rule, originally involving multiplications and divisions, by implementing it using the LAU. By reformulating (16) using LNS, we replace multiplications and divisions, as shown below

$$\begin{aligned}\mu_x' = \mu_x &- \mathrm{Sgn}(\mu_x)\,\mathrm{Sgn}(N_x)\,2^{\left(\log_2|\mu_x| - \log_2|N_x|\right)} \\ &+ \mathrm{Sgn}(N_x)\,\mathrm{Sgn}\left(\sum |f|\right) \\ &\cdot \mathrm{Sgn}(f)\,2^{\left(\log_2|f| - \log_2|N_x| - \log_2(\sum|f|)\right)}\end{aligned} \tag{19}$$

where $\mu_x$ and $N_x$ represent the update parameters for $x \in \{0, 1\}$, and $\mathrm{Sgn}(\cdot)$ denotes the signum function. The feature output iterable is denoted by $f$. During each prediction step, the T3A module accumulates the outputs $f_i$ from all trees, computes their $l_1$-norm, applies the $\mathrm{Sgn}(\cdot)$ function, and passes the resulting values to the LAU block. The updated parameters are then used for the next feature window. After processing each window, the update counter $N_x$ is incremented based on the class (0 or 1) that was updated.

*4) T3A Reuse:* Since only one class parameter update (class 0 or class 1) is active at a time, the same hardware resources can be reused to compute and update the T3A parameters for each feature window. This reuse of hardware enables efficient computation of per-window parameter updates while maintaining minimal hardware complexity.

### F. Multi-Mode Stimulation Control

To enable flexible stimulation strategies for diverse therapeutic needs and symptom profiles, the proposed SoC supports multiple stimulation modes across four independent blocks. The programmable threshold-based stimulation controller is shown in Fig. 16. Stimulation can be triggered based on: 1) per-sample phase or amplitude envelope; 2) windowed biomarkers such as PLV, PAC, or SE; or 3) a combination of both. For example, stimulation may be locked to a specific phase crossing defined by a target threshold ($\mathrm{TH_{SMP}}$), while

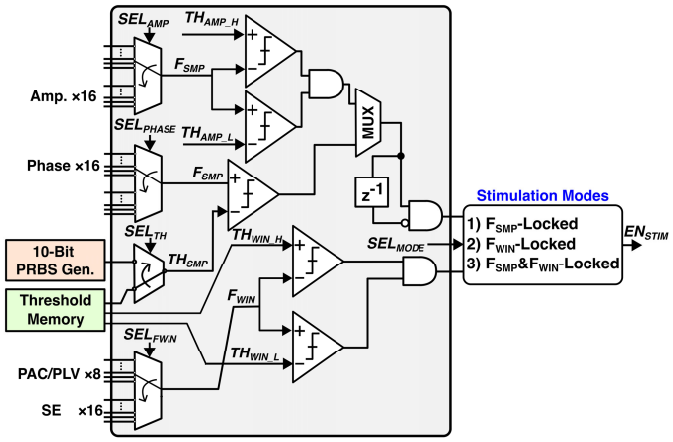



Fig. 16. Block diagram of the programmable threshold-based stimulation controller.

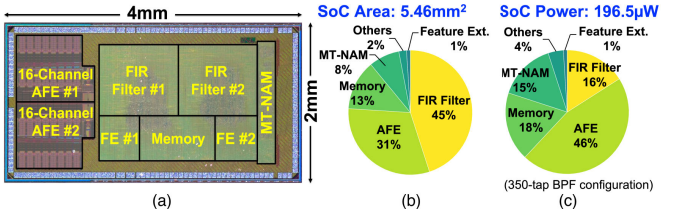



Fig. 17. (a) Chip micrograph, (b) SoC area, and (c) power breakdowns.

ensuring that cross-regional PLV or PAC remains within a predefined therapeutic window bounded by $\mathrm{TH_{WIN,L}}$ and $\mathrm{TH_{WIN,H}}$. For randomized phase-locking, a pseudorandom binary sequence (PRBS) threshold generator is implemented in order to disrupt a target oscillation in the brain. In addition, the stimulation control can be driven by the psychiatric classifier output, using either MT-NAM alone or MT-NAM with T3A, as shown in Fig. 2. The final stimulation decision is user-configurable, allowing flexible integration of classifier outputs and biomarker thresholds to support personalized closed-loop therapy.

## III. Measurement Results

The SoC was fabricated in a TSMC 65-nm 1P9M low-power CMOS process. The chip micrograph and the SoC area and power breakdowns are shown in Fig. 17(a)–(c). The 32-channel SoC occupies an active area of only 5.46 $\mathrm{mm^2}$ (0.17 $\mathrm{mm^2}$/channel), and consumes 196.5 $\mu$W at a 0.85 V digital supply and 1.2-V AFE supply under the 350-tap BPF configuration. Across the supported tap counts, total SoC power ranges from 173 to 196.5 $\mu$W. Fig. 18 shows the total power versus tap count, highlighting the adaptability of the FE engine. This enables tuning of the system based on application requirements, patient-specific conditions, or available power budget while maintaining consistent classification performance.

### A. AFE Characterization

The 32-channel low-noise AFE occupies an area of 1.76 $\mathrm{mm^2}$ (0.055 $\mathrm{mm^2}$/channel) and consumes 94.7 $\mu$W (2.96 $\mu$W/channel). The programmable gain ranges from 53.6

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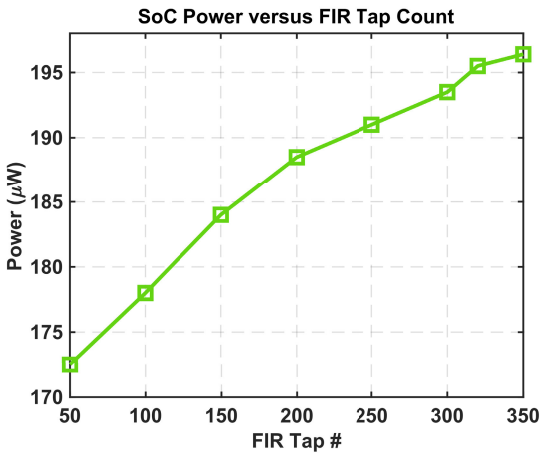



Fig. 18. Total SoC power consumption versus configured tap count.

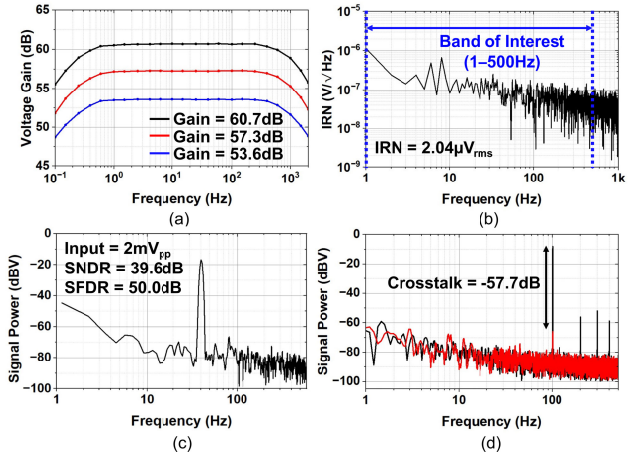



Fig. 19. Measured performance of the 32-channel low-noise AFE: (a) frequency response, (b) IRN spectral density, (c) output power spectrum, and (d) inter-channel crosstalk.

to 60.7 dB, as shown in Fig. 19(a). The measured IRN is 2.04 $\mu V_{rms}$, as shown in Fig. 19(b).

Fig. 19(c) shows the single-tone output power spectrum for a 2 $mV_{pp}$, 40-Hz sine-wave input using a 4096-point FFT. The AFE achieves an SNDR of 39.6 dB and an SFDR of 50.0 dB, primarily limited by the nonlinearity of the $G_m$–$C$ integrator.

Low inter-channel crosstalk is essential in the proposed multi-channel SoC, since cross-channel biomarkers such as PLV and PAC depend on channel independence and can be degraded by artificial correlation arising from channel coupling. Crosstalk was evaluated by driving one channel with a 2 $mV_{pp}$, 100-Hz tone while grounding adjacent channels. The worst case crosstalk between adjacent channels was measured to be −57.7 dB, as shown in Fig. 19(d). The CMRR and power-supply rejection ratio (PSRR) at 50 Hz were measured as 65.6 and 67.6 dB, respectively. The design achieves an input impedance of 153 MΩ at 100 Hz.

### B. Feature Extractor Measurements

The performance of the feature extractor was evaluated by comparing the SoC outputs with a MATLAB reference implementation. Pre-recorded neural signals from the basolateral amygdala (BLA) and infralimbic cortex (IL) of Long-Evans

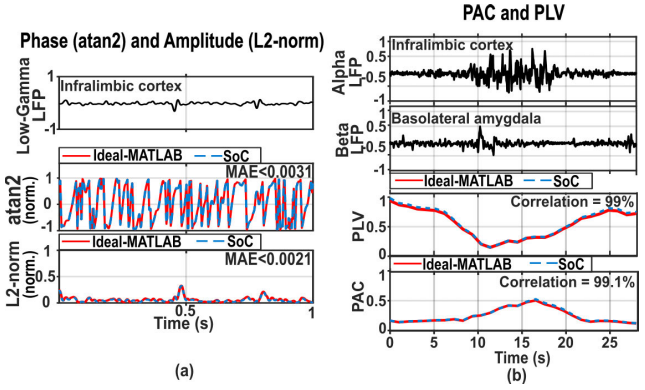



Fig. 20. (a) SoC measurements compared with MATLAB reference for instantaneous phase (atan2) and amplitude envelope ($l_2$-norm), and (b) PAC and PLV computations.

rats were replayed through the AFE using an external DAC. As shown in Fig. 20(a), the extracted instantaneous phase (via atan2) and amplitude envelope (via $l_2$-norm) closely match the MATLAB reference, exhibiting very low mean absolute error (MAE). The SoC's neural connectivity computation is demonstrated in Fig. 20(b), where the computed PLV and PAC show high correlation with MATLAB results. Specifically, PAC/PLV achieve 99% and 99.1% correlation with the ideal MATLAB implementation, respectively, demonstrating robustness to the use of the $\ell_\infty$-norm [i.e., $\max(|X|, |Y|)$] in the aggregation stage, as shown in Fig. 10. In addition, the reduction in PAC correlation observed in [19] (95.4%) is mitigated by employing the proposed $\ell_2$-norm for the amplitude envelope ($A_t$). This confirms the functional accuracy of the proposed log-domain FE architecture described in Section II-B, while significantly reducing hardware complexity.

### C. MT-NAM Classifier Measurements

Analysis of computational requirements shows that distilling the neural network NAM into micro-trees yields substantial hardware efficiency. The original NAM architecture, comprising 48 feature networks with four layers each, requires 519 263 MAC, exponential, and accumulation operations, along with 505 393 parameters (758 kB). In contrast, the depth-4 distilled MT-NAM requires only 287 comparison and accumulation operations and 1488 parameters (2.23 kB), representing a 1809× reduction in computational resources and 340× reduction in memory requirements. A depth-2 tree further reduces complexity to 191 operations and 336 parameters (504 B), achieving a 1505× memory reduction relative to the original NAM.

The on-chip classifier was validated across multiple psychiatrically relevant tasks, including MSIT, ECR, and the public DEAP dataset for valence (V) and arousal (A). Fig. 21 summarizes the performance (sensitivity, specificity, and accuracy) of the quantized MT-NAM classifier, as well as the improvements obtained using the LAU-based T3A update for individual patients. The complete SoC, including the AFE, the FE engine, and the MT-NAM classifier with the T3A update, achieves an overall energy efficiency of 0.197 $\mu$J/class. As shown in Fig. 21, the LAU-based T3A update yields average relative

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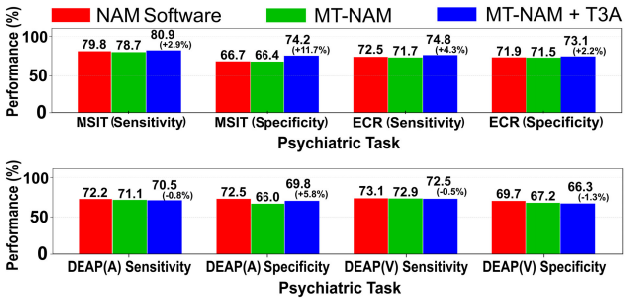



Fig. 21. Mean classification performance of the software-based NAM using neural networks, the offline MT-NAM, and the online MT-NAM with T3A updates, along with the relative performance gain of T3A over the offline MT-NAM, evaluated on the MSIT, ECR, DEAP (A), and DEAP (V) datasets.

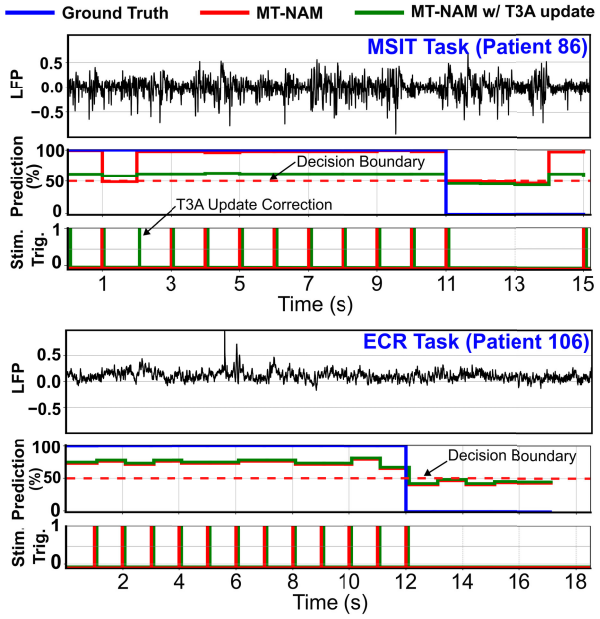



Fig. 22. MT-NAM classifier prediction and decision outputs provided to the stimulation controller.

gains of 1.5% in sensitivity and 4.6% in specificity, with only a 0.3% increase in SoC power.

To support safe closed-loop operation, the neuromodulation trigger is gated by the confidence-based classifier output, as shown in Fig. 22. Two representative patient datasets from the MSIT and ECR datasets are processed through the FE pipeline and subsequently passed to the classifier. The MT-NAM classifier generates event predictions based on the trained parameters, while the T3A module performs online test-time adaptation on the classifier outputs to refine the decision probabilities used for stimulation triggering. The resulting output is thresholded at a fixed 50% decision boundary to generate stimulation triggers. Notably, in cases where the MT-NAM underestimates the event probability, such as around the 2-s time point in the MSIT example, T3A shifts the predicted probability above the decision threshold, thereby correcting a missed stimulation event. All experiments were conducted on clinically annotated datasets with well-defined class balance across patients and experimental conditions.

*T3A Measurements:* The T3A update mechanism relies on high-confidence predictions from MT-NAM and exhibits

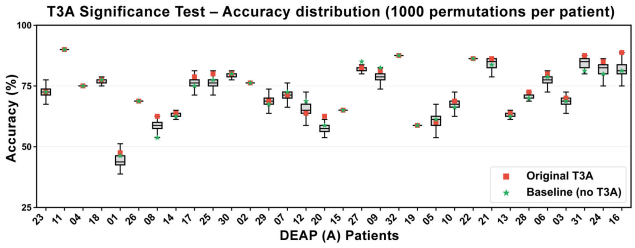



Fig. 23. Accuracy distribution of the T3A online adaptation across 1000 random permutations of high-confidence test sample order for each patient in the DEAP Arousal dataset. Each box represents the interquartile range, with whiskers indicating the full range across permutations. The green star denotes the baseline MT-NAM accuracy without T3A, and the red square indicates the T3A accuracy using the original sample order.

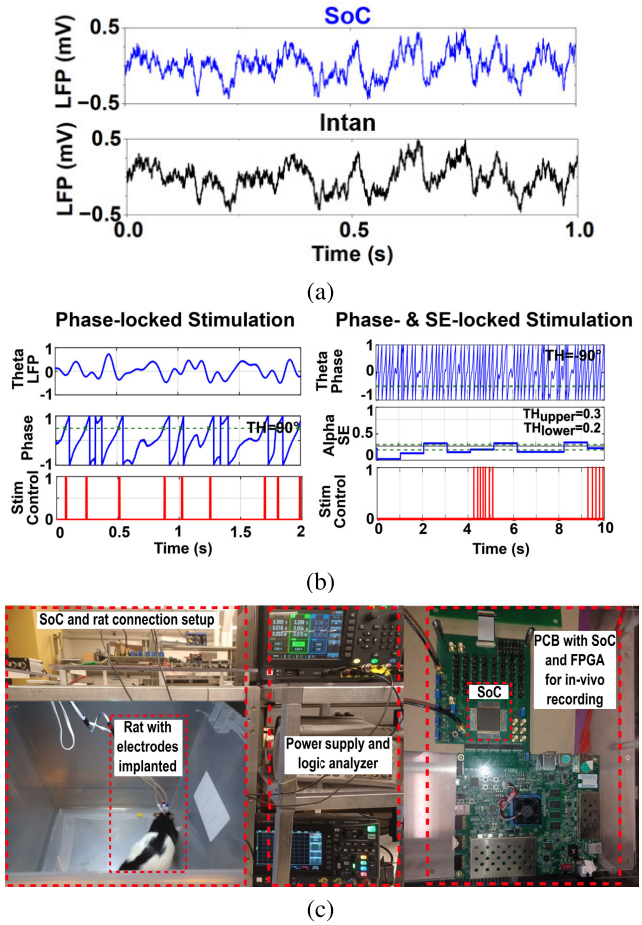



Fig. 24. (a) In vitro LFP recording using the SoC and the Intan C3314, (b) closed-loop stimulation results with triggers based on (left) instantaneous phase only and (right) a combined feature using SE and phase, and (c) in vivo experimental setup.

dataset-dependent behavior. To evaluate robustness to biased drift, we conducted an order-sensitivity stress test on the DEAP arousal dataset by randomly shuffling high-confidence test samples 1000 times per patient and comparing the resulting accuracy distribution to the baseline without T3A (see Fig. 23). This test probes the robustness of T3A under realistic variations in the arrival order of confident samples during online adaptation. For most patients (25 out of 32),

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TABLE I

COMPARISON WITH STATE-OF-THE-ART CONNECTIVITY BIOMARKER EXTRACTION AND EMOTION CLASSIFICATION SOCS

| Parameter | JSSC'18 [29] | TBioCAS'19 [55] | SSCL'23 [19] | JETCAS'19 [56] | TBioCAS'20 [57] | TBioCAS'21 [54] | This Work | | |
|---|---|---|---|---|---|---|---|---|---|
| Process (nm) | 130 | 180 | 65 | 28 | 180 | 180 | 65 | | |
| Power Supply (V) | 1.2 | 0.5 | 1.2/0.85 | 2.5/0.9 | 1 | 1 | 1.2/0.85 | | |
| # of Channels | 32 | 2 | 16 | 6 | 8 | 2 | 32 | | |
| On-Chip AFE | YES | NO | YES | NO | YES | YES | YES | | |
| AFE Gain (dB) | - | - | 53–61 | - | < 80 | 50 – 64 | 54 – 61 | | |
| IRN ($\mu V_{rms}$) | 1.66 (0.1 – 500 Hz) | - | 0.88 (1 – 500 Hz) | - | - | 0.55 (100 Hz) | 2.04 (1 – 500 Hz) | | |
| PAC/PLV Power (μW/Feature) | 3.08 (CFC/PLV) | 0.015 (PLV only) | 1.21 (PAC/PLV) | - | - | - | 0.88* – 1.92† (PAC/PLV) | | |
| Correlation with Ideal Features (%) | - | 90 (PLV) | 98.9 (PLV), 95.4 (PAC) | - | - | - | 96.4*– 99.0† (PLV), 95.7*– 99.1† (PAC) | | |
| Classifier | SVM | - | - | CNN | L-SVM | FC-DNN | MT-NAM w/ T3A Online Update | | |
| Online Adaptation | NO | - | - | NO | NO | NO | YES | | |
| Classification Task | Seizure Detection | - | - | Emotion Recognition | Emotion Recognition | Emotion Recognition | (1) Psychiatric Task (2) Emotion Recognition | | |
| SoC Energy Efficiency (μJ/Class.) | - | - | - | - | 16 | 10.13 | 0.197 | | |
| Psychiatric Dataset (# of Patients) | - | - | - | DEAP (32) | DEAP (32) / SEED (15) | DEAP (32) SEED (15) | MSIT (14) | ECR (12) | DEAP (32) |
| Accuracy (%) | - | - | - | 83.4 (V) 76.7 (A-V) | 73.1 (A) 73.0 (V) / 70.7 | 85.4 | 77.0 | 76.4 | 71.4(A) 71.0(V) |
| Sensitivity (%) | - | - | - | - | - / - | 78.8 | 80.9 | 74.8 | 70.5(A) 72.5(V) |
| Specificity (%) | - | - | - | - | - / - | 89.1 | 74.2 | 73.1 | 69.8(A) 66.3(V) |
| SoC Power (μW) | 674.4 | 0.015 | 60 | 76610 | 2040 | - | 172.5* – 196.5† | | |
| SoC Power/Ch. (μW) | 21.1¶ | 0.008¶ | 3.75 | 12768 | 255 | - | 5.09* – 6.14† | | |
| SoC Area ($mm^2$) | 3.31 | 0.05 | 2.24 | 3.35 | 5.4 | 16 | 5.46 | | |
| SoC Area/Ch. ($mm^2$) | 0.237 | 0.025¶ | 0.14 | 0.56 | 0.675 | 8 | 0.17 | | |

* 50-tap FIR † 350-tap FIR ¶ DBE Only

performance (box plot) remained close to or above baseline, indicating stability to sample-order variations.

For patients with weaker baseline performance (e.g., patient 8), T3A may degrade relative to the baseline when early pseudo-label errors bias the running class prototypes away from the underlying class structure. To mitigate this effect, a higher confidence threshold can be used for patients with larger train-test distribution mismatch, or T3A can be disabled when offline calibration shows that it does not provide a robust accuracy improvement over the MT-NAM baseline. In clinical practice, the T3A enable/disable decision is made per subject using the labeled order-sensitivity analysis, where T3A is enabled only when its accuracy distribution lies above the MT-NAM baseline across sample orderings and is disabled otherwise. Therefore, the proposed T3A module acts as a lightweight and per-subject configurable adaptation mechanism whose effectiveness depends on dataset characteristics, offline calibration quality, and the chosen confidence threshold, rather than providing a guaranteed monotonic improvement.

### *D. In Vitro and In Vivo Measurements*

To validate the recording performance and closed-loop control capability of the proposed SoC, both in vitro and in vivo experiments were conducted. In the in vitro setup, a prerecorded LFP signal was simultaneously applied to the proposed SoC and a commercial neural interface (Intan C3314), ensuring identical input conditions. Each channel was sampled at 4 kS/s and bandpass filtered over 1–500 Hz. As illustrated in Fig. 24(a), the SoC preserved signal fidelity comparable to the commercial system while offering substantially lower power consumption per channel and reduced area overhead.

For in vivo validation, two independent electrode implants targeted the IL and nucleus accumbens shell (NAc-Shell) of Long-Evans rats. Each implant consisted of a twisted pair of 128 $\mu$m Pt-Ir wires, with presurgical electrode impedances of 10–40 kΩ at 1 kHz. A reference screw placed above the cerebellum enabled differential recordings against the brain region of interest. All surgical procedures were performed under isoflurane anesthesia (1%–2.5%) with medical-grade $O_2$ (0.5–1 L/min), adjusted based on body weight and perioperative vital signs.

In vivo recordings using the SoC AFE were performed after a brief postsurgical recovery period. After collection of in vivo neural recordings through the SoC AFE, the performance of the multimode stimulator control was evaluated by replaying this collected neural data directly into the SoC digital back-end (DBE) for further processing. Stimulation control was evaluated using two different strategies: 1) a phase-only threshold crossing and 2) a combined feature using cross-regional spectral energy and instantaneous phase. As shown in Fig. 24(b), both methods successfully triggered stimulation events, with the SE-phase strategy exhibiting improved target selectivity. Stimulation was initiated with low latency, typically within a few milliseconds of biomarker detection, confirming the

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feasibility of real-time neuromodulation using the fabricated SoC.

### E. Comparison With the State of the Art

Table I compares the proposed SoC with recent connectivity-extraction and emotion-classification systems. Unlike prior designs that rely on CORDIC engines [29] or coarse approximations [9], which often incur higher power consumption or reduced accuracy, the proposed LAU-based architecture computes SE, phase, PAC, and PLV with substantially lower energy while closely matching MATLAB reference results. The design achieves a per-feature power consumption of only 0.8–1.92 $\mu$W, depending on tap configuration, providing remarkable accuracy and an excellent energy–accuracy tradeoff. Compared with earlier EEG-based emotion-recognition SoCs, which typically process only 2–8 channels and target high-level affective state classification, the proposed design supports full 32-channel processing and achieves an energy efficiency of 0.197 $\mu$ J/class, substantially lower than prior neural decoding SoCs (e.g., 10.13 $\mu$ J/class in [54]). On iEEG psychiatric datasets, the SoC achieves accuracy, sensitivity, and specificity comparable to software-based studies on these datasets [50] for both MSIT and ECR tasks, setting, to our knowledge, a strong benchmark in energy-efficient psychiatric BCI SoCs for these paradigms.

## IV. Conclusion

This work presents a highly integrated, low-power SoC for neural signal acquisition, FE, and closed-loop control, validated through electrical, in vitro, and in vivo testing. Specifically, we propose a novel LAU for efficient computation of phase and connectivity biomarkers, a hardware-efficient MT-NAM classifier with online T3A adaptation, and their integration into a low-power closed-loop psychiatric BCI. The 32-channel AFE achieves an IRN of 2.04 $\mu$Vrms (1–500 Hz) while consuming only 2.96 $\mu$W per channel, demonstrating high-fidelity LFP sensing. The proposed low-complexity logarithmic feature extractor shows strong agreement with MATLAB reference models when processing pre-recorded LFPs from rat brain regions, while achieving significant reductions in power and area compared to state of the art. The multi-mode stimulation control further demonstrates flexibility for future clinical settings. The on-chip classifier, validated on MSIT, ECR, and DEAP datasets, offers a low-power, adaptive solution for psychiatric state classification. Furthermore, the LAU-based T3A update enhances MT-NAM performance with minimal impact on power consumption. Overall, the proposed SoC advances the state of the art in neural FE and closed-loop capability, representing one of the first integrated BCI platforms designed specifically for psychiatric-state decoding and real-time neuromodulation.

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**Daney Alex** (Student Member, IEEE) received the B.Tech. degree in electronics and communication engineering from Cochin University of Science and Technology, Kochi, Kerala, India, in 2018, and the M.S. (Research) degree in electrical engineering from IIT Palakkad, Palakkad, Kerala, in 2022. He is currently pursuing the Ph.D. degree in electrical engineering with the Swiss Federal Institute of Technology Lausanne (EPFL), Lausanne, Switzerland.

From 2022 to 2023, he was a VLSI Design Engineer with Sinergia Media Labs (Simelabs), Kochi. In 2023, he joined EPFL. His research interests include hardware–algorithm co-design for biomedical and edge-computing applications, energy-efficient digital architectures for signal processing and machine learning, and ASIC implementation and verification.



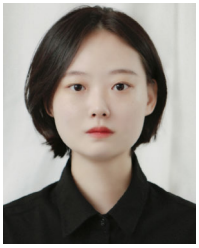
**Amitabh Yadav** (Student Member, IEEE) received the B.Tech. degree in electronics engineering from the University of Petroleum and Energy Studies, Dehradun, India, in 2017, and the M.Sc. degree in computer engineering from Delft University of Technology, Delft, South-Holland, The Netherlands, in 2019.

From 2019 to 2023, he was an Experimental Physicist in the area of quantum computing at Lawrence Berkeley National Laboratory, Berkeley, CA, USA, and UC Berkeley, Berkeley, and later joined as a fellow at European Organization for Nuclear Research (CERN), Geneva, Switzerland. Since 2024, he has been a Doctoral Student at the Integrated Neurotechnologies Laboratory, École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland, where his research focuses on ultralow power digital and mixed-signal integrated circuit design, computing-in-memory and in vivo testing for closed-loop brain–machine interfaces.

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**Jieun Joo** (Student Member, IEEE) received the B.Sc. degree in electronics engineering and the M.Sc. degree in electronic and electrical engineering from Ewha Womans University, Seoul, South Korea, in 2020 and 2022, respectively. She is currently pursuing the Ph.D. degree in microsystems and microelectronics with the Swiss Federal Institute of Technology Lausanne (EPFL), Lausanne, Switzerland.

Her research interests include low-power analog/mixed-signal integrated circuits, data converters, and neural interfaces for biomedical applications. Her current research focuses on energy-efficient architectures for implantable neural recording systems, particularly for signal acquisition and data conversion.



**Uisub Shin** (Student Member, IEEE) received the B.E. degree in electronics engineering from Chungnam National University, Daejeon, South Korea, in 2015, the M.S. degree in electrical engineering from KAIST, Daejeon, in 2017, and the Ph.D. degree in electrical and computer engineering from Cornell University, Ithaca, NY, USA, in 2024.

From 2020 to 2024, he was a Visiting Doctoral Assistant at the École Polytechnique Fédérale de Lausanne (EPFL), Geneva, Switzerland. In 2023, he was an Analog IC Design Intern at Analog Devices, Wilmington, MA, USA. He is currently with Apple, Sunnyvale, CA, USA, developing analog/mixed-signal and digital IPs for high-performance sensing systems. His research interests include analog/mixed-signal ICs, machine learning SoCs, and DSP hardware/algorithms for healthcare and sensing applications.

Dr. Shin was a recipient of the 2023 IEEE SSCS Predoctoral Achievement Award, the 2022 IEEE SSCS-Brain Best Paper Award, and the Kwanjeong Scholarship.



**Jinhan Liu** received the B.Sc. degree in optoelectronics information science and engineering from Beijing Institute of Technology, Beijing, China, in 2017, and the M.Sc. degree in electrical and computer engineering from the University of California, Los Angeles, Los Angeles, CA, USA, in 2019. He is currently pursuing the Ph.D. degree in electrical engineering with the École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland.

He develops robust neural decoding methods for EEG and intracranial recordings. His research interests include machine learning for neural time-series modeling, self-supervised representation learning, emotion recognition, speech brain–computer interfaces, and psychiatric disorders.



**Geoffrey W. Diehl** received the B.A. degree in neuroscience and biochemistry from Oberlin College, Oberlin, OH, USA, and the Ph.D. degree in neurosciences from the University of California, San Diego, La Jolla, CA, USA.

He was a Post-Doctoral Associate at the Department of Neuroscience and subsequently at the Department of Psychiatry, University of Minnesota, Minneapolis, MN, USA. His work has focused on in vivo electrophysiology and neuromodulation in preclinical and translational models. He has most recently used closed-loop electrical stimulation to develop treatments for psychiatric disorders including treatment-resistant depression (TRD) and obsessive-compulsive disorder (OCD).



**Volkan Cevher** (Fellow, IEEE) received the B.Sc. degree (valedictorian) in electrical engineering from Bilkent University, Ankara, Türkey, in 1999, and the Ph.D. degree in electrical and computer engineering from Georgia Institute of Technology, Atlanta, GA, USA, in 2005.

He was a Research Scientist with the University of Maryland, College Park, MD, USA, from 2006 to 2007, and also with Rice University, Houston, TX, USA, from 2008 to 2009. He was also a Faculty Fellow with the Electrical and Computer Engineering Department, Rice University, from 2010 to 2020. He is currently a Full Professor at the Swiss Federal Institute of Technology Lausanne, Lausanne, Switzerland, and an Amazon Scholar. His work has focused on in vivo electrophysiology and neuromodulation in preclinical and translational models. He has most recently used closed-loop electrical stimulation to develop treatments for psychiatric disorders including treatment-resistant depression (TRD) and obsessive-compulsive disorder (OCD). His research interests include machine learning, optimization theory and methods, and automated control.

Dr. Cevher is a fellow of ELLIS. He was a recipient of the ICML AdvML Best Paper Award in 2023, the Google Faculty Research Award in 2018, the IEEE Signal Processing Society Best Paper Award in 2016, the Best Paper Award from CAMSAP in 2015, the Best Paper Award from SPARS in 2009, ERC CG in 2016, and ERC StG in 2011.



**Alik S. Widge** received the M.D. degree from the University of Pittsburgh, Pittsburgh, PA, USA, and the Ph.D. degree in robotics from Carnegie Mellon University, Pittsburgh.

He completed a Psychiatry Residency at the University of Washington, Seattle, WA, USA, and a Fellowship at Massachusetts General Hospital, Boston, MA, USA, and Massachusetts Institute of Technology, Cambridge, MA, USA. He is currently a Brain Stimulation Psychiatrist and a Biomedical Engineer. He is an Associate Professor of psychiatry at the University of Minnesota, Minneapolis, MN, USA, where he directs the Translational NeuroEngineering Laboratory. His research focuses on brain stimulation for severe and treatment-resistant mental illness, with particular emphasis on deep brain stimulation and related implantable technologies.



**Mahsa Shoaran** (Senior Member, IEEE) received the B.Sc. and M.Sc. degrees from the Sharif University of Technology, Tehran, Iran, in 2008 and 2010, respectively, and the Ph.D. degree in electrical engineering from the Swiss Federal Institute of Technology (EPFL), Lausanne, Switzerland, in 2015.

She was a Post-Doctoral Fellow with California Institute of Technology, Pasadena, CA, USA, from 2015 to 2017. From 2017 to 2019, she was an Assistant Professor with the School of Electrical and Computer Engineering, Cornell University, Ithaca, NY, USA. She is currently an Associate Professor with the Institute of Electrical and Micro Engineering and the NeuroX Institute, École Polytechnique Fédérale de Lausanne, Geneva, Switzerland, and the Director of the Integrated Neurotechnologies Laboratory, Geneva. Her research interests include low-power mixed-signal IC design for neural interfaces, machine learning SoCs, and neuromodulation therapies for neurological disorders.

Dr. Shoaran serves on the Technical Program Committee for the IEEE ISSCC, CICC, NER, and the ISSCC SRP Committee. She was a recipient of the 2021 ERC Starting Grant, the 2018 Google AI Faculty Research Award, and two Swiss NSF Postdoctoral Fellowships. She was named a Rising Star in EECS by MIT in 2015. She serves as the Chair in Biomedical Electronics for the IEEE International Conference on Electronics, Circuits and Systems (ICECS) and as an Associate Editor for the IEEE Transactions on Biomedical Circuits and Systems (TBioCAS).

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